

Citywide Street-Light Decomposition with Pole-Inventory Exposure

2026-07-14

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Documentation for `05b_corrected_hawkes_streetlight.py`. This note records what changed in this move, why, the full model, and the validation evidence. It supersedes the street-light-variant description in `05_corrected_hawkes.md` §1 (the “05b” column of its run table): the sl run is no longer a per-district refit of 04 — it is a single citywide fit with a new exposure for the NT equation.

1. Context

Observed 311 illegal-dumping counts conflate two things: how much dumping truly happens, and how willing residents are to call 311. The decomposition (04) separates them with a second report stream that shares the willingness component but carries no dumping signal: street-light-outage reports are filed by the same residents through the same channel, so cross-CBG variation in them pins down the latent reporting propensity w_R ; whatever variation is left in dumping reports after crediting w_R is the reporting-corrected dumping field w_T .

Two weaknesses in the previous implementation motivated this move:

1. Layered pooling. The model was fit independently in each of the 18 planning districts (own β_d , $a_{NT,d}$, $a_{T,d}$, σ 's), and CBG results were only made citywide-comparable through an anchored scalar $a_{T,d} - \beta_d(a_{NT,d} - \bar{a})$ plus a two-layer offset — with a known gotcha (the $\bar{a}$ centring) and a documented negative result (the district-constant offset layer does not land in the Cox-Hawkes a_0).
2. Wrong denominator for outages. Both equations used SafeGraph foot traffic as exposure, so w_R was identified from “outage reports per foot-traffic unit”. But the population at risk of an outage is lights that can break, not visits. A CBG with few poles and lots of foot traffic would look like a low-propensity CBG mechanically.

This move: one citywide fit over all Philadelphia CBGs (no district layer anywhere in the decomposition), with the pole inventory (`data/Street_Poles.geojson`, 203,036 poles) as the NT-equation exposure.

2. Intuition

- With a single fit, $w_{R,i}$ and $w_{T,i}$ are deviations from the citywide level: every CBG is directly comparable to every other, with no within-district-only caveat, no anchor scalar, no centring gotcha. The cross-district level variation formerly absorbed by $a_{NT,d}$ now lives inside w_R itself (so its range widens relative to the old fit — that is signal, not a bug).
- Putting poles in the NT denominator makes the instrument read as it should: “given how many lights this CBG has, how many outage reports does it generate?” Under the identifying assumption below, that ratio is pure reporting propensity.
- The Cox-Hawkes offset collapses from the two-layer $\beta_d(\hat{w}_{R,i} + \hat{a}_{NT,d} - \bar{a})$ to simply $\beta \hat{w}_{R,i}$: with one citywide β and one citywide intercept, there is no district-constant layer left to carry.

2.1 Mechanism: one equation, two unknowns

For each CBG we observe one number — dumping reports — but believe it is the product of two things we cannot see separately:

$$\text{observed dumping reports} = (\text{true dumping}) \times (\text{willingness to report})$$

With only this equation, a CBG with few reports is ambiguous: is it clean, or is it a place where nobody bothers to call 311? Two unknowns per CBG, one measurement — unidentifiable, no matter the model.

The fix is a second measurement that shares exactly one of the unknowns. Under identifying assumption (ii) — lights truly break at the same per-pole rate everywhere —

$$\text{outage reports} = \text{poles} \times (\text{citywide breakage constant}) \times (\text{willingness to report})$$

contains no CBG-specific “true rate” unknown: breakage is a constant, poles are data, and the only CBG-varying term left is willingness. This equation is therefore solvable on its own — a CBG’s outage reports per pole, relative to the city’s, is its reporting propensity w_R . Plugged back into the dumping equation, only one unknown remains, and the leftover is w_T . The whole design is: use the stream with one unknown to solve for the shared unknown, then subtract it out of the stream with two.

In log form (what the code fits, §4), the anatomy is visible directly:

$$\begin{aligned} \log \mathbb{E}[NT_i] &= \log P_i + a_{NT} + w_{R,i} && \leftarrow \text{one unknown per CBG: solvable} \\ \log \mathbb{E}[T_i] &= \log E_i + a_T + \beta w_{R,i} + w_{T,i} && \leftarrow w_R \text{ known from above; leftover} = w_T \end{aligned}$$

Why β ? Willingness to report a broken streetlight and willingness to report dumping are related but not identical behaviors; β is the exchange rate between them, fit from how strongly outage-per-pole rates and dumping rates co-move across the 1,335 CBGs. This run found $\beta = 0.58$: a CBG filing $2\times$ the average outage reports tends to file about $2^{0.58} \approx 1.5\times$ the dumping reports at equal true dumping.

2.2 A worked example (this run’s fitted numbers)

Citywide constants from the fit: outage rate $e^{\hat{a}_{NT}} = e^{-1.08} \approx 0.34$ reports per pole over 2021–25; dumping rate $e^{\hat{a}_T} \approx 0.0052$ per foot-traffic unit; $\beta = 0.58$. Take two hypothetical CBGs that look identical in the dumping data — same foot traffic (10,000 stops), same 52 dumping reports, both exactly at the citywide average rate — differing only in outages:

	poles	outage reports	outages/pole vs city	w_R	$e^{\beta w_R}$	predicted dumping	observed	w_T
CBG A (“chatty”)	100	60	$1.76\times$	$+0.57$	1.39	≈ 72	52	-0.33
CBG B (“quiet”)	100	17	$0.50\times$	-0.69	0.67	≈ 35	52	$+0.40$

The arithmetic, spelled out for CBG A: outages per pole $60/100 = 0.60$ against the citywide 0.34 gives the ratio 1.76, so $w_R = \ln 1.76 = +0.57$; the propensity multiplier entering the dumping equation is $e^{\beta w_R} = e^{0.58 \times 0.57} = 1.39$; propensity alone therefore predicts $52 \times 1.39 \approx 72$ dumping reports, and the shortfall gives $w_T = \ln(52/72) = -0.33$. (CBG B mirrors it: $w_R = \ln 0.50 = -0.69$, prediction ≈ 35 , $w_T = \ln(52/35) = +0.40$.)

Same raw dumping counts, opposite conclusions. A’s residents call 311 about everything (76% above average on lights); given that chattiness we would expect ~ 72 dumping reports and see only 52 — A is truly cleaner than

it looks. B's residents rarely call (half the average); given that silence, 52 reports is a lot — B is truly dirtier. Raw counts rank A and B equal; the corrected field w_T ranks B well above A. This reversal is the entire purpose of the model.

2.3 Why a joint Bayesian fit instead of that arithmetic

The table above is the deterministic skeleton, and for large-count CBGs the model does essentially that. The Poisson + priors machinery earns its keep at the edges:

- Small counts. A CBG with 20 poles and 2 outage reports has a wildly noisy ratio; the prior $w_R \sim \mathcal{N}(0, \sigma_R^2)$ shrinks it toward the citywide average instead of taking $2/20$ literally (the bend in `sl_ppc_fit.png`).
- Simultaneity. β , the citywide constants, and all $\sim 2,670$ latent values are estimated together, so uncertainty in one propagates into the others (per-CBG w_{R_sd} , w_{T_sd} in the output).
- No division by zero. Everything is counts and offsets; zero-report CBGs are handled naturally.

One caveat the trick cannot escape: it corrects relative geography, not the absolute amount of unreported dumping — a citywide uniform under-reporting factor is invisible to any within-city comparison.

3. Identifying assumptions

- Reporting behavior is uniform within a CBG — the same latent propensity drives both outage reports and dumping reports from a CBG's residents.
- The true physical rate of streetlight breakage per pole is uniform across the city. Under (ii), variation in reported outages per pole reflects reporting propensity, not real breakage — which is what lets outage reports proxy the reporting rate we divide out of dumping reports.

The PSIP threat to (ii). The Philadelphia Street Lighting Improvement Program replaced old bulbs with LEDs, which fail far less. The pole file shows the conversion landed mostly inside the study window's tail: of $\sim 117k$ dated relights (`light_date`), only $\sim 21k$ predate 2024 vs $\sim 96k$ in 2024–2026, and the per-CBG LED share spans roughly 45%→75% (p10→p90). So during 2024–25 true breakage fell non-uniformly across CBGs — exactly the kind of real-rate variation assumption (ii) rules out. Two robustness levers address it (§6): a pre-conversion time slice (NT counts from 2021–2023 only, when the stock was still mostly old bulbs) and an LED-share sensitivity check (does w_R correlate with conversion geography?).

Design decisions fixed in advance: the denominator is all poles regardless of owner (residents report outages without knowing whether Streets or PECO owns the pole, and we have no 311 routing data); `pole_date` is not a temporal filter (nearly all poles predate the window); the denominator is pole count, not luminaire count (`n_lumin` is missing for all PECO poles and contains zeros/junk).

4. Model

One fit over all CBGs i with $E_i > 0$ and $P_i > 0$ (counts pooled over 2021–2025):

$$NT_i \sim \text{Poisson}(P_i e^{a_{NT} + w_{R,i}}) \quad (\text{outage equation})$$

$$T_i \sim \text{Poisson}(E_i e^{a_T + \beta w_{R,i} + w_{T,i}}) \quad (\text{dumping equation})$$

- P_i = street poles in CBG i (exposure, coefficient fixed at 1)
- E_i = SafeGraph average monthly stop counts (`all_loc_avg_s_cnt`), unchanged from 04 — the dumping equation still measures dumping per unit of activity
- NT_i = Street Light Outage 311 reports (single category; the 5-category spec-A instrument lives in 04's main run)
- T_i = Illegal Dumping 311 reports

Priors (identical to 04; non-centred parameterisation for NUTS):

parameter	prior	role / interpretation
a_{NT}	$\mathcal{N}(0, 5^2)$	citywide log outage-report rate per pole
a_T	$\mathcal{N}(0, 5^2)$	citywide log dumping-report rate per unit foot traffic
β	$\text{LogNormal}(0, 0.5)$	loading of reporting propensity into dumping reports (one citywide value; formerly 18 β_d)
σ_R, σ_T	$\text{HalfNormal}(1)$	spread of the two latent fields
$w_{R,i} = \sigma_R z_{R,i}$ $z_{R,i} \sim \mathcal{N}(0, 1)$	—	CBG i 's log reporting-propensity deviation from the citywide level
$w_{T,i} = \sigma_T z_{T,i}$ $z_{T,i} \sim \mathcal{N}(0, 1)$	—	CBG i 's log reporting-corrected dumping deviation — the target quantity

Retired relative to the per-district spec: $\beta_d, a_{NT,d}, a_{T,d}, \sigma_{\cdot,d}$ (18 of each $\rightarrow 1$), the anchor $\bar{a} = \log(\sum NT / \sum E)$, and the deterministic corrected_level _{d} = $a_{T,d} - \beta_d(a_{NT,d} - \bar{a})$. With one fit, cross-CBG comparability is built in and the corrected citywide level is just a_T .

Cox-Hawkes offset. The reporting-thinning offset entering the background log-intensity of the per-district Cox-Hawkes fits (05's machinery, unchanged) becomes

$$\text{offset}_i = \hat{\beta} \hat{w}_{R,i} \quad (\text{plug-in posterior means})$$

CBGs without a decomposition estimate get 0 (the citywide mean). The district-constant layer of the old offset is gone by construction; the corrected_level column consumed by 05's a_0 -comparison figure is retired (returned as NaN, which makes that figure skip itself — it was a documented negative result anyway).

Fitting. NUTS (numpyro), target_accept = 0.9, 1000 warmup + 1000 samples, 1 chain, seed 0 — 04's settings, one run over ~1,325 CBGs (~2,650 latent z 's) instead of 18 small runs. Point-process side: bstpp Cox-Hawkes per district for Central and University Southwest, three specs (baseline / offset_fixed / offset_free), exactly as in 05.

5. The pole inventory (EDA facts)

data/Street_Poles.geojson, 203,036 poles, EPSG:4326 $\rightarrow$ 26918.

- Coverage: 202,386 poles fall inside a 2023 CBG (650 outside the layer). Poles per CBG: median 131 (p5 54, p95 312). Only 3 CBGs have zero poles (excluded from the fit); 9 have <10 (kept — the Normal prior shrinks them).
- Owner: Streets 169,971 (83.7%), PECO 23,463 (11.6%), blank 6,878, plus 13 minor owners (PennDOT, Private, CCD, ...). PECO poles carry no attribute data: 98.7% tap_id = 0, 99.9% psip_status = "N/A", bulb_type UNKNOWN, nlumin/lum_size/height missing. Spatially PECO is a small minority nearly everywhere — median CBG share 9%, p90 22%, only 4 CBGs above 50% — so PECO heterogeneity cannot drive cross-CBG variation. All poles stay in the denominator per the design decision.
- Attributes: type is dominated by WP (103k) / SNP (24k) / AEL (18k); nlumin/lum_size/height are 33.8% missing; light_date 42.2% missing; block/plate 91% missing. nlumin has 6,913 zeros and junk values (3614, 320) — reinforcing the pole-count (not luminaire-count) denominator.
- PSIP: psip_status COMPLETED 123,513 (60.8%), N/A 76,991 (mostly PECO + minor owners), PENDING/OTHER ~1.3k each. bulb_type LED 123,726 (60.9%). Dated relights: 2023 $\approx$ 11k, 2024 $\approx$ 65k, 2025 $\approx$ 30k — see §3.

Outputs: poles_cbg.geojson (per-CBG counts and LED/PECO/pre-2024 shares), pole_eda_*.csv tables, and figures pole_map_per_cbg.png, pole_map_led_share.png, pole_hist_per_cbg.png, pole_light_date_hist.png.

6. Validation and robustness results

Run of 2026-07-13. All correlations are in validation_summary.csv; scatters under output/figures/corrected_hawkes

Model fit. 1,335 / 1,338 CBGs (excluded: 3 zero-pole CBGs, all in the 9892xx special-use tract; none lacked exposure). $T = 116,751$ dumping reports, $NT = 75,956$ outage reports. NUTS: ~ 10 s, 0 divergences. $\hat{\beta} = 0.579$ [0.470, 0.686] (94% CI) — comfortably inside the old per-district range and below 1, echoing 04's finding that propensity carries into dumping reports with a loading well under proportional. $\hat{a}_{NT} = -1.077$ (≈ 0.34 outage reports per pole over the window), $\hat{a}_T = -5.264$, $\hat{\sigma}_R = 0.620$, $\hat{\sigma}_T = 1.488$. PPC in sl_ppc_fit.png; citywide maps sl_map_w_R.png, sl_map_w_T.png.

check	what it tells us	pearson	spearman	n
w_R citywide vs archived district-wise fit	how much the refactor changed the field (old fit was within-district centred, foot-traffic exposure)	0.68	0.63	1,335
w_R vs log outages-per-pole	mechanical sanity check — should be ≈ 1	0.99	1.00	1,324
w_R full window vs 2021–23 NT window	PSIP time-slice robustness — high corr means assumption (ii) survives the LED conversion	0.95	0.95	1,335
w_R vs LED share	PSIP sensitivity	0.28	0.22	1,335
w_R vs pre-2024-relit share	PSIP sensitivity	-0.15	-0.19	1,335
log poles vs log foot traffic	how different the two denominators are	0.46	0.51	1,335

Decision rule (pre-registered in the script): if $|\text{spearman}(w_R, \text{LED share})| > 0.3$, the 2021–23 NT window becomes the headline spec. Result: $0.22 \leq 0.3$, and the pre-conversion time slice reproduces the full-window field at $r = 0.95$ (its $\hat{\beta} = 0.588 \approx 0.579$) — the full-window NT stands as the headline spec; assumption (ii) is robust to the PSIP conversion. The moderate 0.63 correlation with the old district-wise fit is expected, not alarming: the citywide field now carries the cross-district level variation the old fit absorbed into its 18 intercepts, and the denominator changed (poles vs foot traffic — themselves only 0.51 correlated).

Cox-Hawkes (Central + University Southwest $\times$ baseline / offset_fixed / offset_free; 6/6 fits successful; offset range fed to the model $[-1.26, +1.12]$, vs $[-7.8, +3.1]$ under the old two-layer offset):

district	spec	a_0	α	excitation share	γ (94% CI)
Central	baseline	-1.28	0.20	~ 0	—
Central	offset_fixed	-1.32	0.20	~ 0	—
Central	offset_free	-1.32	0.20	~ 0	1.22 [1.15, 1.29]

district	spec	a_0	α	excitation share	γ (94% CI)
University Southwest	baseline	-1.50	0.20	~0	—
University Southwest	offset_fixed	-1.51	0.20	~0	—
University Southwest	offset_free	-1.50	0.20	~0	-0.11 [-0.27, 0.05]

The headline is Central's γ : under the old district-layer, foot-traffic-based offset it was 0.63 (already the highest after conditioning on the GP and covariates, which absorb shared signal and push γ down); with the citywide, pole-denominated offset the point process now estimates γ at or above 1 — the spatiotemporal pattern of dumping reports, data 04 never saw, independently endorses the correction as full thinning. University Southwest stays ≈ 0 as before (University City's dumping pattern carries little within-district reporting signal; its offset field is comparatively flat). Both districts remain essentially free of self-excitation, consistent with the earlier runs. The `ch_scatter_a0_vs_corrected_level` figure is retired with `corrected_level` itself (it documented a negative result).

7. Figure guide

All under `output/figures/corrected_hawkes_streetlight/`, in pipeline order.

Pole inventory EDA (Step 0)

pole_light_date_hist.png — PSIP relighting dates vs the study window. Bar chart of `light_date` years, 2021–2025 shaded. The visual statement of the assumption-(ii) threat: relights are negligible until 2023, then ~11k (2023), ~65k (2024), ~30k (2025) — the LED conversion happened inside the window's tail, which is why the robustness checks in §6 exist.

pole_map_per_cbg.png / pole_hist_per_cbg.png — the NT denominator. Poles per CBG as choropleth + histogram. Coverage is dense and even (median 131/CBG, no holes) — what makes poles a usable exposure. The 3 gray zero-pole CBGs (airport/navy-yard tract) are the excluded ones.

pole_map_led_share.png — the geography of the PSIP threat. Share of each CBG's poles with `bulb_type = LED`. If this map mirrored the w_R map, "propensity" would partly be "conversion timing". It is patchy — low-LED pockets in parts of North Philly, Grays Ferry, near the airport — but does not mirror w_R (spearman 0.22 < 0.3 threshold).

Citywide decomposition (Step 1)

sl_map_w_R.png — reporting propensity, one citywide scale. Log deviation from the citywide outage-per-pole rate ($+1 \approx 2.7\times$ the citywide reporting rate, $-1 \approx 0.37\times$). Unlike the old within-district maps, colors are comparable anywhere on the map. High propensity through Center City, the South Philly rowhouse core, scattered Northeast pockets; systematically depressed (-1.5 to -2) across the Lower Southwest and the industrial riverfronts.

sl_map_w_T.png — the target: reporting-corrected dumping, citywide comparable. After stripping willingness-to-report, the elevated band runs across North Philadelphia and Kensington, with pockets in West and Southwest Philly; the far Northeast is uniformly low. Put this next to the raw dumping-rate map to show what the correction changes.

sl_ppc_fit.png — posterior-predictive check. Observed vs fitted (log-log), one panel per equation; both hug the 45° line over 3+ orders of magnitude. The bend at the bottom-left (fitted > observed for CBGs with 0–3 reports) is partial pooling shrinking tiny-count CBGs toward the citywide mean — intended, not pathological (0 divergences).

Proxy validation & PSIP robustness

val_scatter_wR_vs_outage_per_pole.png — mechanical sanity check. w_R vs raw log(outages/pole); spearman ≈ 1.0 by construction (w_R is the shrunk, intercept-removed version of this ratio). Departures from the line are the low-count CBGs being shrunk.

val_scatter_wR_full_vs_pre24.png — the key robustness figure. w_R with NT from 2021–2023 only (pole stock still mostly old bulbs) vs w_R from the full window: a tight 45° band, spearman = 0.945, no curvature or fanning from LED-converted CBGs. Even though true breakage fell non-uniformly in 2024–25, it barely moves the propensity field — assumption (ii) survives.

val_scatter_wR_vs_led_share.png — no-refit sensitivity. Diffuse cloud, spearman 0.22. Had this been strong, “high propensity” would really have meant “not yet converted”. It isn’t → full-window spec stands (§6 decision rule).

val_scatter_wR_city_vs_district.png — what the refactor did. Archived district-wise w_R (foot-traffic exposure, within-district centred; note its extreme $-7..+5$ range from the old exposure artifacts) vs the new citywide estimate (compressed to about ± 2). $r = 0.63$: clearly related, deliberately not identical — the citywide fit re-injects the cross-district level differences the old fit hid in its 18 intercepts, and the denominator changed (poles vs foot traffic, themselves only 0.51 correlated).

Cox-Hawkes step (Central + University Southwest)

ch_map_offset_cbg.png — the offset fed to the point process. Red = above-average propensity (intensity thinned down), blue = suppressed reporting (scaled up); range $[-1.26, +1.12]$ vs the old offset’s $[-7.8, +3.1]$. Known cosmetic caveat: the title still prints the old two-layer formula $\beta_d(\hat{w}_R + \hat{a}_{NT,d} - \bar{a})$ — it is hardcoded in 05 (untouched); the data plotted is the new $\beta \hat{w}_R$.

ch_map_f_xy_norm_baseline.png / **ch_map_f_xy_norm_offset_fixed.png**. The fitted GP spatial surface f_{xy} aggregated to CBGs, min-max normalized within each district, before/after correction — where within each district the background intensity concentrates (03a’s caveat applies: compare within a district only).

ch_map_f_xy_norm_delta.png — corrected – baseline. Red = a CBG becomes a relatively stronger hotspot once reporting is corrected; blue = it was partly a reporting artifact. The core of Center City turns blue (high-propensity residents inflated its apparent intensity) while Central’s northern and southern fringes turn red; University Southwest shifts more mildly the same way. Magnitudes are small (± 0.1 on a 0–1 scale): the correction reshapes hotspots at the margin, it does not redraw them.

ch_map_a_0_offset_fixed.png. Background intercept a_0 under the main spec (Central -1.32 , USW -1.51). Record-keeping with two districts; per the §8 negative result in 05_corrected_hawkes.md, do not read a_0 as a cross-district corrected level.

ch_scatter_excitation_shift.png. α and excitation share, baseline vs corrected, one point per district. Both districts sit on the diagonal at excitation ≈ 0 : no meaningful self-excitation, unchanged by the correction (unlike the 18-district spec-A run, where correction lowered excitation).

ch_dumbbell_cov_weights.png. Posterior-mean covariate weights, baseline vs offset_fixed, averaged over the two districts. Covariates whose dots move were partly proxying for reporting propensity in the baseline.

ch_forest_offset_coef.png — the validation headliner. offset_free lets the data scale the offset by γ ; $\gamma = 1$ (dashed line) = the correction is exactly right. Central: $\gamma = 1.22 [1.15, 1.29]$ — at or above full thinning, up from 0.63 under the old offset: the spatiotemporal dumping pattern (data the decomposition never saw) independently endorses the pole-based correction. University Southwest: $\gamma = -0.11 [-0.27, 0.05] \approx 0$ — its point pattern carries little within-district reporting signal (its offset field is comparatively flat), same as before.

8. Files

file	status	role
05b_corrected_hawkes_streetlights.py	newly written	pole EDA + join, citywide two-exposure fit, PSIP robustness, offset override, Cox-Hawkes driver count/exposure builders reused as modules; its own per-district spec-A run is unaffected Cox-Hawkes batch reused; its build_offset is replaced in memory by the citywide version hard constraint
04_reporting_decomposition.py	untouched	
05_corrected_hawkes_by_district.py	untouched	
cox_hawkes_offset.py, ~/BSTPP, 03/03a	untouched	
output/corrected_hawkes_streetlights.py	newly written	
.../reporting_decomp_city_summary.csv	newly written	citywide per-CBG w_R/w_T (+ w_R_mean_pre24, pole columns) two rows: full-window and pre-24 citywide scalars
.../reporting_decomp*_districts.csv	archived	
		the pre-refactor district-wise outputs, kept for the comparison in §6